Experiment: 5

Simulation and Throughput Analysis of LTE-Advanced Carrier Aggregation

Aim

To simulate an LTE-Advanced wireless link using carrier aggregation and analyze the effect

of aggregating multiple component carriers on available transmission bandwidth and

estimated throughput using MATLAB.

Objectives

• To model an LTE-Advanced link using one component carrier.

• To extend the model to use multiple aggregated component carriers.

• To compare estimated throughput for different carrier bandwidths.

• To explain how carrier aggregation increases available transmission bandwidth.

Software Required

• MATLAB

Theory

Carrier aggregation is an LTE-Advanced technique in which multiple component carriers are

combined so that a user can access a larger effective transmission bandwidth. A single

component carrier provides a limited bandwidth, whereas aggregation of two or more carriers

increases the total usable bandwidth. In a simplified throughput model, estimated throughput

can be related to aggregated bandwidth and spectral efficiency. The experiment therefore

demonstrates the principle of bandwidth aggregation and its impact on achievable data rate.

Expected Outputs

• Plot/table showing throughput for a single component carrier.

• Comparison of one, two and three aggregated component carriers.

• Throughput versus aggregated bandwidth graph.

• Observation that aggregated bandwidth increases the estimated throughput under the same

spectral-efficiency assumption.

Objective – 1: Model One Component Carrier

Define a single LTE-Advanced component carrier and estimate its throughput for a selected

bandwidth and spectral efficiency.

Procedure

1. Open MATLAB and create a new script.

2. Define the component-carrier bandwidth, spectral efficiency and number of component

carriers.

3. Calculate the single-carrier throughput using bandwidth × spectral efficiency.

4. Display the bandwidth and estimated throughput.

5. Plot the single-carrier result.

MATLAB Program

```
clc; clear; close all;

BW1 = 20; % Component carrier bandwidth (MHz)

SE = 4; % Spectral efficiency (bits/s/Hz)

Ncc = 1; % Number of component carriers

BWagg = BW1*Ncc;

Throughput = BWagg*SE; % Mbps for MHz × bits/s/Hz

disp('Single Component Carrier')

disp(['Bandwidth = ',num2str(BWagg),' MHz'])

disp(['Estimated Throughput = ',num2str(Throughput),' Mbps'])

figure

bar(Ncc,Throughput)
```

grid on

xlabel('Number of Component Carriers')

ylabel('Estimated Throughput (Mbps)')

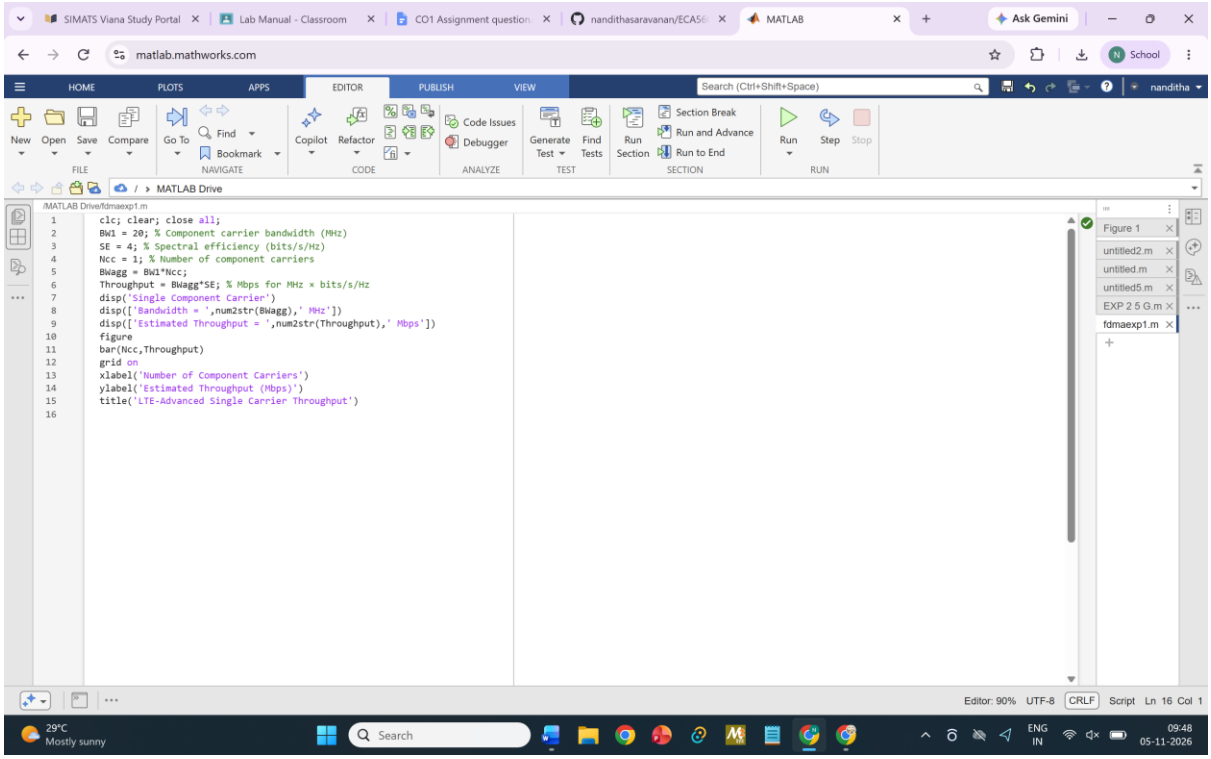
title('LTE-Advanced Single Carrier Throughput')

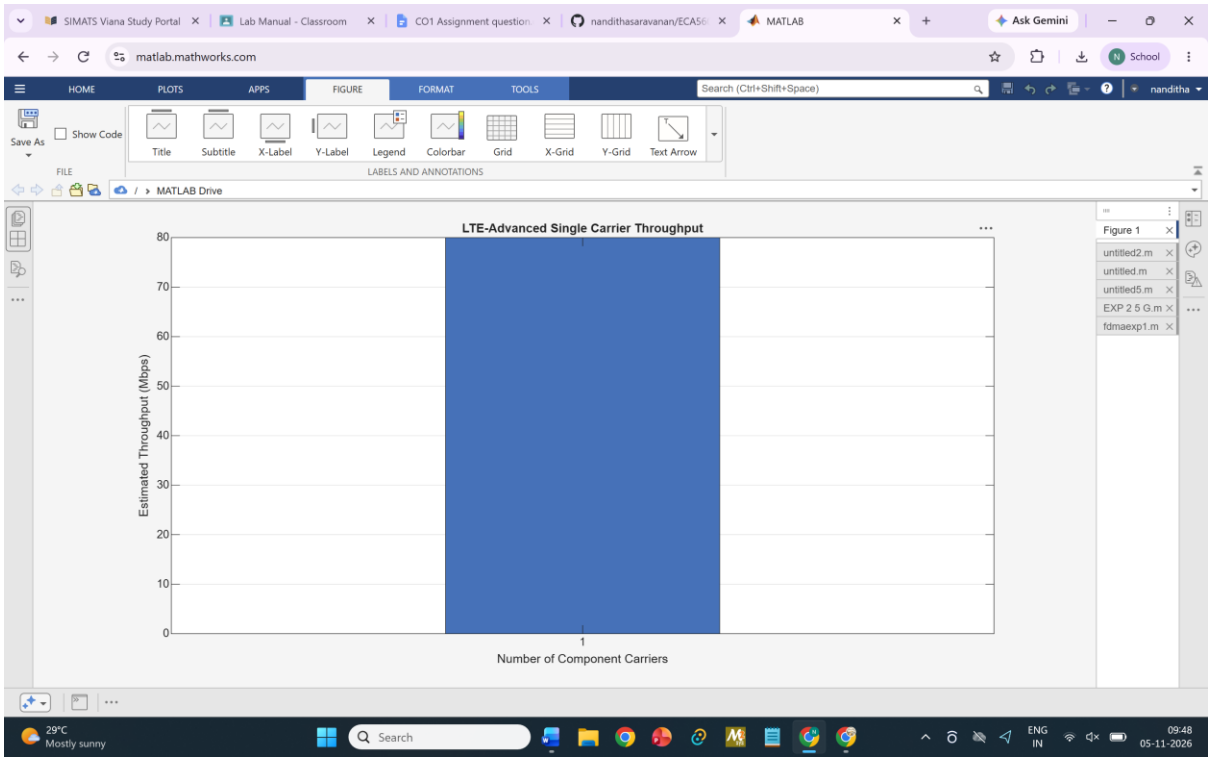
Expected Output: A graph showing the estimated throughput for one 20 MHz component

carrier. With the selected 4 bits/s/Hz simplified assumption, the estimated throughput is 80

Mbps.

Output:





Objective – 2: Extend to Multiple Aggregated Component Carriers

Aggregate multiple component carriers and determine the resulting total bandwidth and estimated throughput.

Procedure

6. Define one, two and three component-carrier configurations.

7. Calculate total aggregated bandwidth for each configuration.

8. Estimate throughput using the same spectral-efficiency value.

9. Plot throughput for the different aggregation configurations.

10. Compare the results and explain the increase.

MATLAB Program

```
clc; clear; close all;

BWcc = 20; % MHz per component carrier

SE = 4; % bits/s/Hz

CC = [1 2 3]; % Number of component carriers

AggBW = BWcc .* CC; % MHz

Throughput = AggBW .* SE; % Mbps
```

disp('CC Aggregated BW (MHz) Estimated Throughput (Mbps)')

disp([CC' AggBW' Throughput'])

figure

bar(CC,Throughput)

grid on

xlabel('Number of Aggregated Component Carriers')

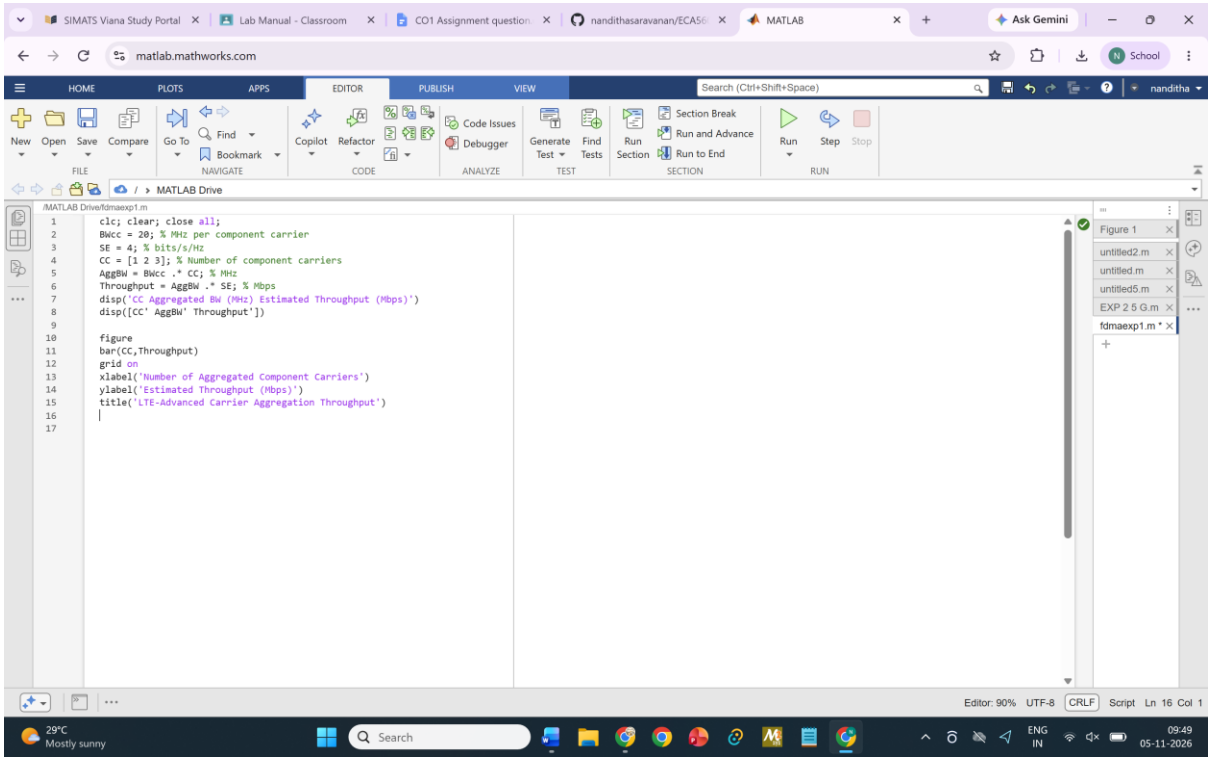
ylabel('Estimated Throughput (Mbps)')

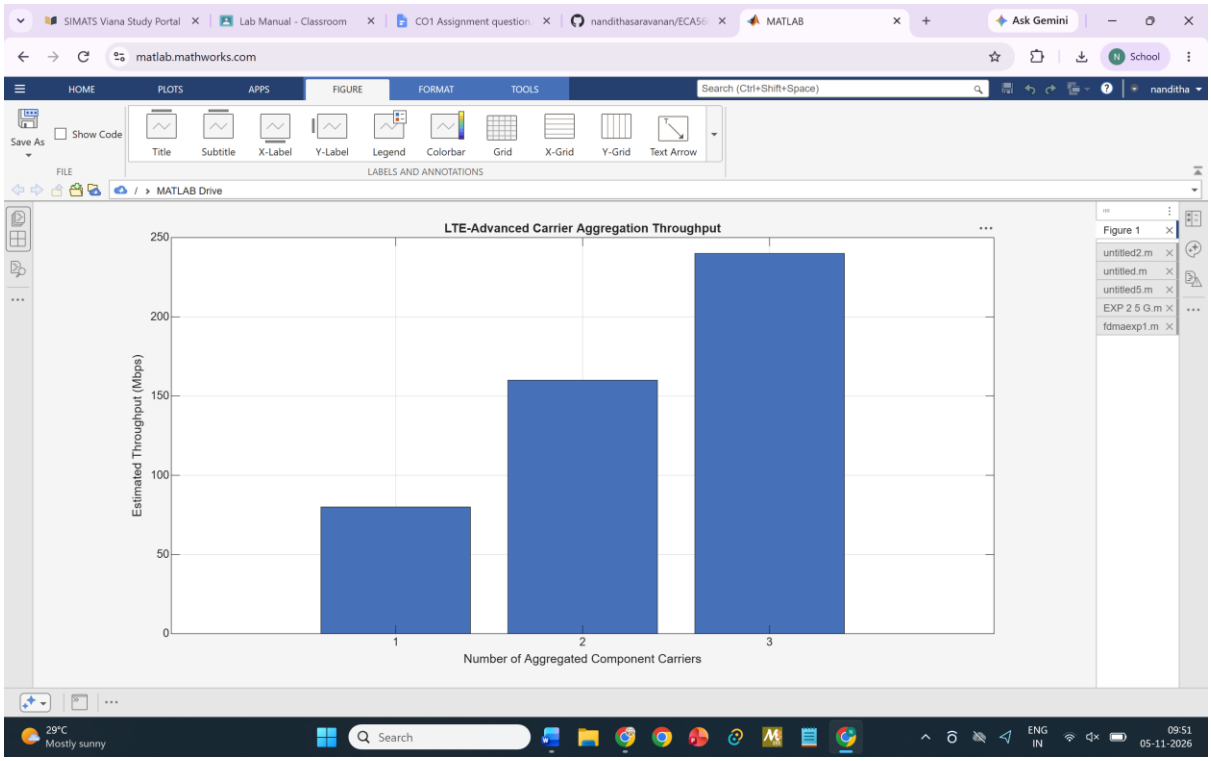
title('LTE-Advanced Carrier Aggregation Throughput')

Expected Output: The graph increases with the number of aggregated component carriers. For

the selected parameters, 1, 2 and 3 carriers provide 80, 160 and 240 Mbps respectively.

Output:





Objective – 3: Compare Estimated Throughput for Different Carrier Bandwidths

Procedure

1. Define several component-carrier bandwidths.

2. Calculate throughput for each bandwidth using the selected spectral efficiency.

3. Compare the throughput values.

4. Plot bandwidth against estimated throughput.

5. Identify the bandwidth configuration that provides the highest estimated throughput.

MATLAB Program

```
clc; clear; close all;

BW = [5 10 15 20]; % Carrier bandwidths (MHz)

SE = 4; % bits/s/Hz

TP = BW .* SE; % Estimated throughput (Mbps)

disp('Bandwidth (MHz) Throughput (Mbps)')

disp([BW' TP'])


figure
```

plot(BW,TP,'o-','LineWidth',2)

grid on

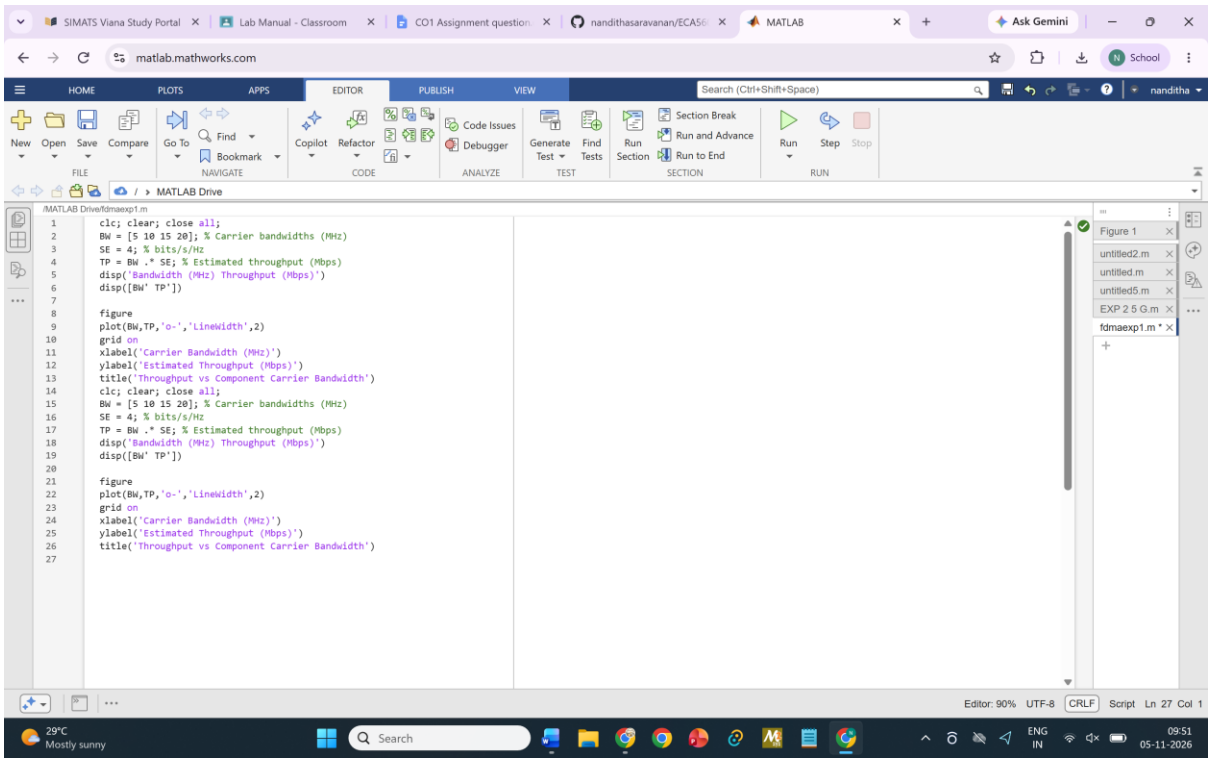
xlabel('Carrier Bandwidth (MHz)')

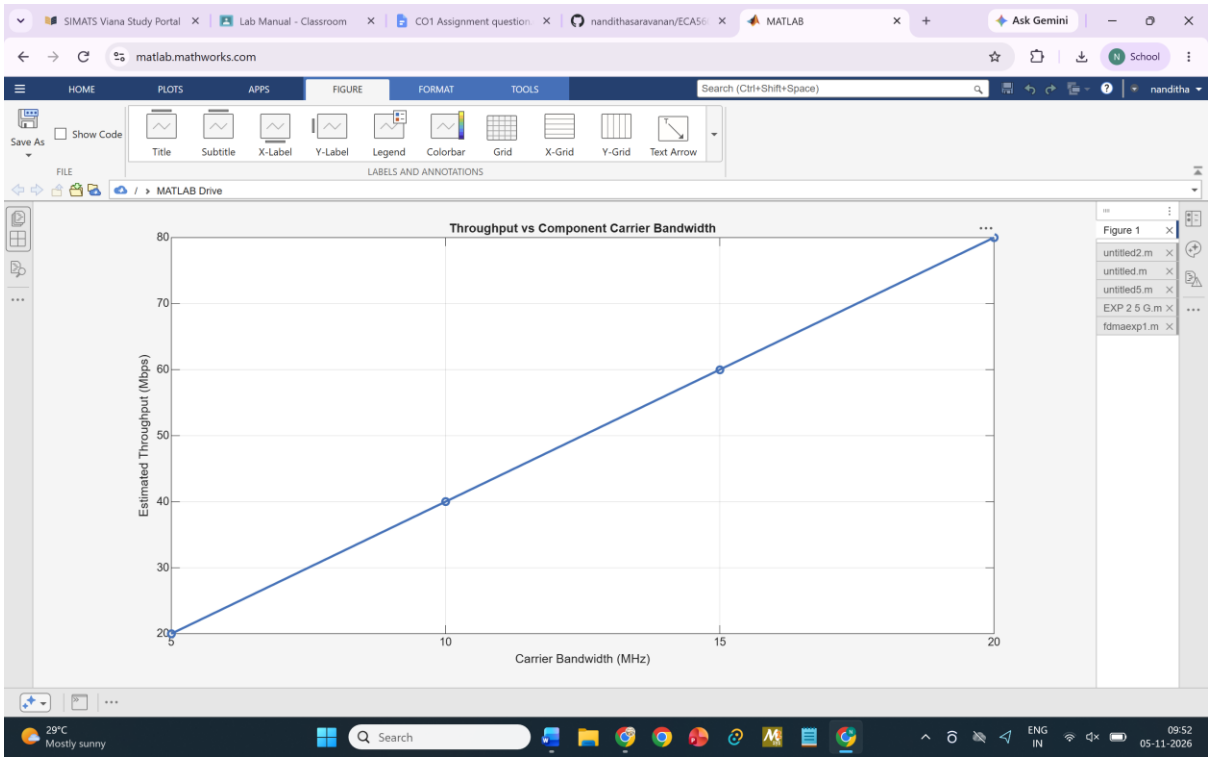
ylabel('Estimated Throughput (Mbps)')

title('Throughput vs Component Carrier Bandwidth')

Expected Output: Throughput increases as the component-carrier bandwidth increases.

Output:





Objective – 4: Explain the Effect of Carrier Aggregation

The total available bandwidth is obtained by combining the component carriers. Under the

simplified constant spectral-efficiency model, throughput increases approximately in

proportion to the aggregated bandwidth. In a practical LTE-Advanced system, the actual

throughput is also influenced by modulation, coding, channel quality, control overhead,

scheduling and radio conditions.

Result

1. The LTE-Advanced link was successfully modeled using a single component carrier.

2. Multiple component carriers were aggregated and the resulting bandwidth and estimated

throughput were calculated.

3. Throughput was compared for different carrier bandwidths.

4. The simulation demonstrated that carrier aggregation increases available transmission

bandwidth and can increase achievable throughput.

Discussion

Carrier aggregation improves LTE-Advanced capacity by allowing multiple component carriers to be treated as a larger effective transmission resource. The MATLAB results demonstrate the basic relationship between aggregated bandwidth and estimated throughput.

Increasing the number of carriers increases the total bandwidth available to the user. This concept provides the basis for higher data rates and improved spectrum utilization in LTE-Advanced systems and is an important foundation for understanding wider-bandwidth techniques used in 5G.